

Tilt-Based Control Barrier Function for Target Visibility

A camera-plane QP over the body tilt via the rotational interaction matrix

Soft–Precise–Landing

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Scope. Visibility is an *attitude* constraint — the body tilt moves the field of view — so we guarantee it with a small QP (a clamp) on the desired tilt, on the *real* camera plane. The coupling between tilt and feature is the **rotational part of the IBVS interaction matrix**, L_ω , evaluated at the measured image point: it is exact, *depth-independent*, and needs no de-rotation / virtual frame. The barrier uses image features only — scale- and depth-free (no β , no optic flow, no disturbance term). Notation follows the manuscript.

Notation

Symbol	Meaning
${}^c\hat{\mathbf{r}}_i, {}^c\hat{\mathbf{r}} = (x, y)$	camera image feature points / centroid, tangent units, per axis k
$L_\omega({}^c\hat{\mathbf{r}})$	rotational interaction matrix (roll/pitch columns), $\partial {}^c\dot{\hat{\mathbf{r}}} / \partial \boldsymbol{\omega}_{\text{rp}}$
$\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the <i>rotation-axis</i> vector)
$\boldsymbol{\theta}, \theta_d, \theta^*$	body tilt = <i>lean-direction</i> vector ($\boldsymbol{\theta}_d = R_z(-\psi) \mathcal{I}_{a_{xy}/a_z}$); desired, safe
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	lean→rotation map: $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (lean $+x$ is a rotation about $+y$)
$\widetilde{L}_\omega = L_\omega M$	effective tilt coupling (orthogonal M , so row norms are preserved)
$\mathbf{d}$	exogenous (translation) feature drift, $\mathbf{d} = {}^c\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\varphi_{\text{max}} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
$\delta, \dot{\delta}$	primary-marker half-spread (loom rate)
θ_{cap}	deliverable tilt cap (60°); τ drift look-ahead; h barrier

1 Barrier on the real camera plane

Per image axis k ,

$$h_k = \varphi_{\text{max},k} - |{}^c\hat{r}_k| - \delta_k. \quad (1)$$

$h_k \geq 0 \iff$ the marker (outermost feature point) is on the sensor. ${}^c\hat{\mathbf{r}}$ is the *measured* camera image point — nothing is de-rotated.

2 Tilt→feature coupling: the rotational interaction matrix

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point ${}^c\hat{\mathbf{r}} = (x, y)$:

$${}^c\dot{\hat{\mathbf{r}}} = L_\omega({}^c\hat{\mathbf{r}})\boldsymbol{\omega}_{\text{rp}} + \mathbf{d}, \quad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}, \quad (2)$$

where $\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$ is the body roll/pitch *rate* and $\mathbf{d}$ is the exogenous (translation / target) drift. L_ω contains **no depth** — the tilt coupling needs no β and no virtual frame. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\mathbf{d} = {}^c\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega\boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$ (using the true gyro rate $\boldsymbol{\omega}^{\text{curr}}$), so it carries only the genuine target motion, not our own tilting.

Lean vector vs. rotation axis. The QP variable $\boldsymbol{\theta}$ is the *lean-direction* vector — the horizontal direction of the desired thrust/acceleration, $\boldsymbol{\theta}_d = R_z(-\psi)^T a_{xy}/a_z$ — not the body angular-rate vector that drives (2). A lean toward $+x$ is produced by a rotation about the $+y$ axis, so the two differ by a fixed 90° rotation,

$$\boldsymbol{\omega} = M\boldsymbol{\theta}, \quad M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \quad (3)$$

The feature therefore moves with the *effective* coupling $\widetilde{L}_\omega \equiv L_\omega M$ (validated against an independent pinhole+attitude camera model: $< 8\%$ over the near-hover envelope). For a finite tilt change the exact map is the rotation homography of the measured point; the first-order form

$${}^c\hat{\mathbf{r}}(\boldsymbol{\theta}) \approx {}^c\hat{\mathbf{r}} + \widetilde{L}_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}}) = {}^c\hat{\mathbf{r}} + L_\omega M(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}}) \quad (4)$$

is the local linearization, accurate over one control step. (The drift term keeps $L_\omega\boldsymbol{\omega}^{\text{curr}}$ unchanged — $\boldsymbol{\omega}^{\text{curr}}$ is a measured *rate*, already in the rotation-axis frame, so no M is applied there.)

3 The visibility filter: a QP over the body tilt

With m_k the predicted margin (see Sect. 4),

$$\boxed{\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_d\|^2 \text{ s.t. } |{}^c\hat{\mathbf{r}}_k + [\widetilde{L}_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})]_k + \tau d_k| \leq m_k}, \quad \widetilde{L}_\omega = L_\omega M \quad (5)$$

A small 2-variable QP (the two tilt axes), with ≤ 2 binding rows, solved by a few projection iterations onto the rows of $\widetilde{L}_\omega$. $\widetilde{L}_\omega = L_\omega M$ is near-diagonal for a centred marker (at ${}^c\hat{\mathbf{r}} = 0$, $L_\omega = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = M^{-1}$, so $\widetilde{L}_\omega = I$ — a clean per-axis clamp, closed-form there); the off-diagonal carries the genuine roll↔pitch cross-coupling near the FoV edge, which the iterations resolve.

- **Directional** — only the outward binding row is active; inward (recovery) and tangential tilts are free. No strangling.
- **Input-aware** — after the box projection converges, apply the radial saturation $\boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}^* \cdot \min(1, \theta_{\text{cap}}/\|\boldsymbol{\theta}^*\|)$ so the commanded tilt is always within what the SO(3) inner loop can hold. Applying the cap *inside* the projection loop would interfere with the box constraint's convergence; the post-solution saturation preserves the deliverability bound without creating infeasibility.

- **Forward-invariant** — the look-ahead shifts the anchor by $\tau \mathbf{d}$, so the QP accounts for centroid drift over the next τ s in both phases. In **Phase 2**, the predicted margin is additionally shrunk by $\tau \dot{\delta}_{\text{eff}}$, forcing the inward tilt before the fill rate consumes the remaining headroom ($\dot{h} \geq -\alpha(h)$). In **Phase 1** ($m_k = \varphi_{\text{max}}$), the $\tau \dot{\delta}$ term is excluded by design: the marker is allowed to grow, so only the centroid-drift look-ahead $\tau \mathbf{d}$ is active. $\tau = 0$ gives the static clamp.

4 Two-phase δ (central-marker convention)

The designated *central* marker is the target. The predicted margin m_k is phase-dependent:

Phase 1 — central marker decodes. $\delta_k = 0$, so $m_k = \varphi_{\text{max},k}$ (full half-FoV). The barrier is centroid-only: the QP box spans the entire image plane, and only fires if the *centroid* drifts toward the edge. The central marker is deliberately allowed to grow and overflow as the UAV closes in — its visibility *is* the guarantee; using its large half-extent would fire the barrier early. $\dot{\delta}$ is tracked but excluded from m_k in Phase 1.

Phase 2 — central marker overflows (decode-fail, hysteresis-gated against flicker $\Rightarrow$ touchdown range). δ_{eff} **ramps in** over a few frames from the visible-*set* half-extent ($\frac{1}{2}\text{ptp}_k$ over ALL visible markers' corners, so $|\mathcal{C}\hat{\mathbf{r}}| + \delta_{\text{eff}}$ is the true outermost feature — not one marker):

$$m_k = \varphi_{\text{max},k} - \delta_{\text{eff},k} - \tau \dot{\delta}_{\text{eff},k}.$$

The ramp (not a step) keeps h from dropping discontinuously at engagement, but clamp the ramp gain so δ_{eff} never *lags* the measured $\frac{1}{2}\text{ptp}$ — a ramp slower than the fill under-protects. Within Phase 2 the visible markers themselves leave (a mild inner sawtooth); $\dot{\delta}$ is reset across those switches.

Implementation: hysteresis threshold = 3 consecutive decode-fails before Phase 2 activates; ramp completes over 5 control cycles (≈ 120 ms at 42 Hz). Both $\delta_{\text{eff}} = \alpha \delta_{\text{ref}}$ and $\dot{\delta}_{\text{eff}} = \alpha \dot{\delta}_{\text{ref}}$ are scaled by the same ramp factor $\alpha \in [0, 1]$, where δ_{ref} and $\dot{\delta}_{\text{ref}}$ are the half-extent and loom rate measured at the last successful decode.

5 Feasibility floor and hand-off

With θ_{cap} applied outside the QP, the box constraint (5) is **infeasible** iff $m_k < 0$ for some k (the marker fills the frame within τ). In Phase 1, $m_k = \varphi_{\text{max},k} > 0$ always, so the QP is **always feasible** while the central marker decodes.

The perception-floor hand-off is therefore triggered by *central-marker overflow* (Phase 2 activation, hysteresis-gated), not by QP infeasibility. On Phase 2 entry, or if $m_k < 0$ under Phase 2 margins, raise the perception-floor flag and hand off to the texture-free ring flow.

6 Assumptions (and the practical challenge each carries)

1. **The coupling $\widetilde{L}_\omega = L_\omega M$ and its sign convention are calibrated.** The QP variable is the lean vector $\boldsymbol{\theta}$, but L_ω couples the rotation-axis rate; the fixed 90° map $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (3) bridges them, so the QP uses $\widetilde{L}_\omega = L_\omega M$. Any residual mount-dependent sign of L_ω is a

one-time check: command a small tilt and confirm $\Delta^{\mathcal{C}\hat{\mathbf{r}}}$ matches $\widetilde{L}_\omega$ in sign before trusting the barrier.

2. **The filtered tilt is realized** (SO(3) loop tracks $\boldsymbol{\theta}^*$). *Challenge:* attitude bandwidth $\Rightarrow$ lag; θ_{cap} bounds magnitude, not delay.
3. **The drift $\mathbf{d}$ is the translation part** (${}^{\mathcal{C}}\dot{\mathbf{r}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$) and $\dot{\delta}$ the loom. *Challenge:* finite differences $\Rightarrow$ filter + binding-axis hysteresis; the disturbance enters as *measured* motion — no separate $\mathbf{d}_h$ term. The homography centroid (hence ${}^{\mathcal{C}}\dot{\mathbf{r}}_{\text{obs}}$) must stay *smooth across marker-set changes* (the homography re-fits on a different visible subset) or $\mathbf{d}$ spikes — filter / hold a stable subset across a switch.
4. **First-order L_ω over the cycle.** Exact is the rotation homography of ${}^{\mathcal{C}}\hat{\mathbf{r}}$; the $L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})$ linearization is accurate over one control step. At $\theta_{\text{cap}} = 60^\circ$ the integral (homography) is the safe choice for the boundary $\boldsymbol{\theta}$.
5. θ_{cap} **is the true deliverable tilt** (thrust unsaturated).

7 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the box (5) contain only the measured image point ${}^{\mathcal{C}}\hat{\mathbf{r}}$, δ , $\varphi_{\text{max}} = \mathcal{R}/(2f)$, the body rate, and the tilt. L_ω **is depth-independent by construction** (no $\mathcal{V}z_t$, no β), so the whole formulation is manifestly scale- and depth-free. The only metric quantity upstream is the geometric tracker’s R_d (a controller output).

8 Implementation (per control cycle)

1. (*once*) Calibrate L_ω ’s sign / axis convention (Assumption 1).
2. Measure ${}^{\mathcal{C}}\hat{\mathbf{r}}$ (board centroid, homography). Advance the two-phase δ state: if central marker decodes, reset Phase 2 counters and set $\delta_{\text{eff}} = 0$; otherwise increment the decode-fail counter, and once the hysteresis threshold is reached, ramp δ_{eff} toward the visible markers’ $\frac{1}{2}\text{ptp}$.
3. Evaluate $L_\omega({}^{\mathcal{C}}\hat{\mathbf{r}})$; form the drift $\mathbf{d} = {}^{\mathcal{C}}\dot{\mathbf{r}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$ and $\dot{\delta}_{\text{eff}}$ (filtered; zero in Phase 1).
4. Two-phase predicted margin: Phase 1: $m_k = \varphi_{\text{max},k}$; Phase 2: $m_k = \varphi_{\text{max},k} - \delta_{\text{eff},k} - \tau \dot{\delta}_{\text{eff},k}$. If $m_k \leq 0$ for any k , raise the perception-floor flag and hand off to the ring estimator. Otherwise build the FoV-constraint QP (5).
5. $\boldsymbol{\theta}_d = R_z(-\psi) \mathcal{I} a_{d,xy} / a_z$ (the lean vector from the desired lateral accel); solve (5) with $\widetilde{L}_\omega = L_\omega M \rightarrow \boldsymbol{\theta}^*$ (pure FoV-box alternating projection, always converges when $m_k > 0$). Apply post-QP deliverability cap: $\boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}^* \cdot \min(1, \theta_{\text{cap}} / \|\boldsymbol{\theta}^*\|)$.
6. **Apply $\boldsymbol{\theta}^*$ directly as the desired attitude.** Build the desired body- z axis from the safe tilt, $\mathbf{r}_{d3} = [-R_z(\psi)\boldsymbol{\theta}^*, 1] / \|[\cdot, 1]\|$, and pass R_d to the SO(3) inner loop, taking the thrust *magnitude* from the descent loop. This avoids re-encoding $\boldsymbol{\theta}^*$ as $a_{xy} = a_z R_z(\psi)\boldsymbol{\theta}^*$ and low-pass-filtering it: a_z cancels in $-\mathcal{I}a / \|\mathcal{I}a\|$, so the direction is identical, but the accel-path LPF (time constant τ_{ia}) would smear the barrier’s hard bound by several frames — the guarantee would then not land on the commanded attitude. The direct map makes the deliverable-tilt guarantee exact.